

# Aarya Venkat, PhD

aaryakat@outlook.com — LinkedIn/aaryavenkat

GitHub/arikat — GitLab/avenkat2

## PROFESSIONAL SUMMARY

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Computational Protein Engineer with over 9 years of experience driving ML-powered enzyme design and high-throughput variant prediction. Proven success optimizing enzymes for a variety of KPIs by merging structural bioinformatics with automated MLOps pipelines. Adept at pushing cross-functional projects to commercial success in fast-paced biotech environments.

## EDUCATION

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**University of Georgia** — PhD Computational Biochemistry — Dec 2023

*Achievements: 17 publications, ARCS Foundation Scholar, Three Minute Thesis award*

**University of California, San Diego** — MS Chemistry — June 2017

## TECHNICAL SKILLS

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**Programming:** Python, Bash, Git, Snowflake, Docker, Kubernetes, Argo Workflows, Nextflow, AWS/GCP

**Bioinformatics:** Metagenomic Sourcing, Sequence analysis, Intron/Exon analysis, NGS, Clustering

**Cheminformatics:** RDKit, QSAR, xTB, CREST, molecular docking, virtual screening

**Computational Chemistry:** Molecular Dynamics, DFT, MDAnalysis, Amber, GROMACS, Desmond, PyMol, Schrodinger Maestro, forcefield parameterization, BioEmu, protein-ligand interaction analysis

**Protein Design:** Inverse Folding (ProteinMPNN, LigandMPNN), EVCouplings, Mutational inference (ThermoMPNN, ESM2), co-folding, Predictive modeling, de novo protein design (RFDiffusion),

**AI/ML:** PyTorch, Explainable AI (SHAP), linear/logistic regression, Protein Language Models (ESM2), fine-tuning (PEFT, LoRA), Dataset preparation, training ML models for protein optimization

**Research & Soft Skills:** Technical Writing, Data Visualization and Presentation, Public Speaking

## PROFESSIONAL EXPERIENCE

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**Ginkgo Bioworks, Inc. — Scientist III**

*Dec 2023 – Present — Boston, MA*

- Designed T7 RNA Polymerase as Ginkgo Bioworks' first **commercial enzyme product** with 99.6% reduction (over wt T7) of dsRNA, a toxic byproduct with immunogenic effects.
- Lead over 35 protein engineering campaigns utilizing AI/ML, molecular modeling, and evolutionary models to enhance specificity, thermostability, and activity for high-budget contracts.
- Optimized enzymes in ML-guided active-learning campaigns to reach up to **45x higher activity over wt** in three rounds, with tiny library sizes of 200-350 variants per round.
- Utilized and combined metagenomic sourcing with structure-prediction tools (ESMFold, Chai, Boltz) to achieve protein discovery objectives, identifying proteins with **10x higher activity, thermostability at 90C, halotolerant (stable under 1M salt)**, along with identification of **enzymes that perform novel chemistry**, previously unseen in literature.
- Serve as Subject Matter Expert and Tech Liaison for AI and computational biology, leading customer-facing meetings to communicate research benefits, risks, and strategic roadmaps.
- Architected **Kubernetes** pipelines using **Argo** and **Docker** for atomic workflows like **RFDiffusion-AA**, proteinMPNN, as well as larger meta-workflows created to automate protein design.
- Developed **PyTorch** deep learning models fine-tuned via **LoRA** from ESM2 to build family-specific enzyme models for Ginkgo Bioworks' 'enzyme custom plates' service.

## University of Georgia — Graduate Research Assistant

Aug 2017 – Dec 2023 — Athens, GA

- Authored 17 peer-reviewed manuscripts, contributing to writing, experimental design, data analysis and visualization, and successful funding acquisitions.
- Conducted protein structure-function analyses including molecular modeling and dynamics, intron-exon analysis, structure-based analyses, phylogeny, and quantum chemistry to study enzyme function.
- Worked on protein kinases, glycosyltransferases, and ion channels.
- Managed lab operations like data storage infrastructure, procurement, and protocol documentation.

## PRODUCTS

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**T7 ClearCirc RNA Polymerase (Inventor):** Directed the protein engineering of specific RNAP mutations reducing toxic dsRNA byproduct by 99.6%, significantly reducing immunogenicity and streamlining downstream purification for in vitro and in vivo applications. Technology serves as the core of a commercial drop-in replacement kit for high-fidelity circular RNA synthesis.

## SELECTED PUBLICATIONS

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Total: 23 — Citations: 444 — H-index: 12 — co-first author<sup>†</sup>

**Venkat, A.** (2026). GapClean, a tool for clarifying protein sequence alignments. *In Prep.*

Fang, R. et al. (2026). Glydentify: An explainable deep learning platform for glycosyltransferase donor substrate prediction. *bioRxiv*.

Zhang, J. et al. (2025). ICEPIC: A Toolkit to Discover Ice Binding Proteins from Sequence. *bioRxiv*.

**Venkat, A.** et al. (2023). Mechanistic and evolutionary insights into isoform-specific ‘supercharging’ in DCLK family kinases. *eLife*.

Zhou, Z. et al. (2023). xAI meets biology: A comprehensive review of explainable ai in bioinformatics applications. *arXiv*.

Meng, Y. et al. (2023). Phosphorylation-dependent pseudokinase domain dimerization drives full-length MLKL oligomerization. *Nature Communications*.

Aceil, J.<sup>†</sup>, **Venkat, A.**<sup>†</sup>, et al. (2023). Prevalence and homology of the pneumococcal serine-rich repeat protein at the global scale. *Microbiology Spectrum*.

**Venkat, A.**, et al., (2022). Modularity of the hydrophobic core and evolution of functional diversity in fold A glycosyltransferases. *Journal of Biological Chemistry*.

Taujale, R., **Venkat, A.**, et al. (2020) Deep evolutionary analysis reveals the design principles of fold A glycosyltransferases. *eLife*.

## PATENTS

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**Venkat, A.**, Castro, Y., and Manning, T. Using Computational QSAR Methods to Propose a New Group of Antibiotics for Dental Applications, U.S. Provisional Patent Appl. 62/359,638, July 22, 2016.

## AWARDS

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2023 — ASBMB Conference Travel Award

2021 — ARCS Foundation Scholar

2021 — Three Minute Thesis Award

2021 & 2022 — Glycobiology Poster Award & Travel Award